Convergence guarantees for numerical algorithms for time-inconsistent equilibrium transport problems

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Website #15 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

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Problem

Fix a finite horizon T. Given path laws μ on $X_{1:T}$ and ν on $Y_{1:T}$, and the bicausal feasible set $\Pi_{\mathrm{bc}}(\mu, \nu)$, the task is to design numerical schemes that compute a subgame-perfect equilibrium bicausal transport plan for time-inconsistent objectives, and to prove convergence of the algorithmic outputs to an equilibrium in a natural topology.

I prove a rigorous finite-state theorem. It covers:

- nonlinear objectives of the form $G\big(\int \psi \, d\pi\big)$ (take ψ as a terminal/full-path statistic), and
- state-dependent preferences, by allowing the current self's nonlinear aggregator to depend on the realized history.

The theorem gives:

1. an exact backward-sweep algorithm that computes a strong equilibrium profile of bicausal kernels, and
2. a convergence theorem for inexact backward sweeps: every limit point is an equilibrium, and under a local uniqueness assumption the whole sequence converges in kernel ℓ^1 and hence in total variation of plans.

Alternative readings considered

The prompt admits a very broad reading: arbitrary Polish spaces and all three mechanisms of time inconsistency (general nonseparable f-divergence, nonlinear $G(\int h \, d\pi)$, and state-dependent preferences). I first aimed for that reading, but the strongest theorem I could make fully rigorous and self-contained is the fully discrete finite-state case with a general history-dependent nonlinear continuation criterion. This is strictly stronger than the merely additive/state-dependent special case, and it directly covers the canonical nonlinear example $G(\int h \, d\pi)$.

Related literature found

- **Bayraktar–Han (2025),** ***Equilibrium Transport with Time-Inconsistent Costs*** [PARTIAL]. This is the closest source I found: the abstract states existence/uniqueness results for equilibrium transport under bicausal constraints for nonlinear, state-dependent, and f-divergence costs, plus numerical studies. From the accessible abstract alone I could not audit an iterate-convergence theorem of the finite-state numerical kind proved below, so I did not import any proof from it.
- **Backhoff, Beiglböck, Lin, Zalashko (2017),** ***Causal Transport in Discrete Time and Applications*** [TOOL]. Useful for orientation: it shows that causal transport admits a dynamic-programming structure in the time-consistent case.

- **Moulos (2020/2021),** ***Bicausal Optimal Transport for Markov Chains via Dynamic Programming*** [PARTIAL]. Useful nearby solved problem: time-consistent bicausal OT for Markov chains, with value-iteration flavor. It does not address time inconsistency.

- **Bayraktar–Han (2025),** ***Fitted Value Iteration Methods for Bicausal Optimal Transport*** [PARTIAL]. Useful computational near-neighbor: approximate dynamic programming for time-consistent bicausal OT. It does not cover equilibrium transport with time-inconsistent objectives.

- **Björk–Murgoci (2014),** ***A Theory of Markovian Time-Inconsistent Stochastic Control in Discrete Time*** [BACKGROUND]. This supplies the equilibrium/consistent-planning viewpoint: time inconsistency is treated game-theoretically via subgame-perfect equilibria.

- **Bayraktar–Han (2023),** ***Short Communication: Existence of Markov Equilibrium Control in Discrete Time*** [PARTIAL]. Useful background on discrete-time equilibrium existence, but without the transport/bicausal structure used here.

I did *not* use any external proof as a logical step below; the proof is self-contained.

Proof sketch

In the finite-state case, a bicausal plan can be parameterized by *local transport kernels*: at each history $h_{t-1} = (x_{1:t-1}, y_{1:t-1})$, one chooses a coupling of the prescribed one-step conditional marginals $\mu_t(\cdot \mid x_{1:t-1})$ and $\nu_t(\cdot \mid y_{1:t-1})$. Thus the feasible set becomes a finite product of transport polytopes. This turns the equilibrium problem into a finite extensive-form intrapersonal game.

The crucial observation is that for the nonlinear continuation criteria considered here, time inconsistency does *not* force one to remember the whole future plan. It is enough to carry two continuation objects:

$$A_t(h_{t-1}) = \text{expected future linear cost from } t, \qquad B_t(h_{t-1}) = \text{expected future summary vector from } t.$$

Future selves are fixed when the current self deviates. Hence the current self's one-step optimization problem depends on the current kernel only through the induced distribution of the immediate successor history h_t, and therefore through the affine expressions involving A_{t+1} and B_{t+1}. This yields an exact backward recursion.

For numerical analysis, I allow ε_n-accurate local solves. Because the profile space is compact in the finite case, every output sequence has convergent subsequences. A continuity argument shows that every cluster point satisfies the exact stagewise best-response inequalities, hence is an equilibrium. If each local exact problem has a unique minimizer, then the backward-recursion profile is unique; therefore every subsequence has the same limit, and the whole sequence converges. Since the induced plan is a continuous polynomial function of the kernels, kernel convergence implies total-variation convergence of the transport plans.

Main result

From now on, assume every X_t and Y_t is finite. Write

$$H_t := X_{1:t} \times Y_{1:t}, \qquad H_0 := \{\varnothing\}.$$

For $h_{t-1} = (x_{1:t-1}, y_{1:t-1}) \in H_{t-1}$, fix one-step conditionals

$$\mu_t(\cdot \mid x_{1:t-1}), \qquad \nu_t(\cdot \mid y_{1:t-1}),$$

defined arbitrarily on null histories if needed, and define the transport polytope

$$\Gamma_t(h_{t-1}) := \left\{ \gamma \in \mathcal{P}(X_t \times Y_t) : \sum_{y_t} \gamma(x_t, y_t) = \mu_t(x_t \mid x_{1:t-1}),\ \sum_{x_t} \gamma(x_t, y_t) = \nu_t(y_t \mid y_{1:t-1}) \right\}.$$

A *kernel profile* is $\gamma = (\gamma_t)_{t=1}^T$ with $\gamma_t(\cdot, \cdot \mid h_{t-1}) \in \Gamma_t(h_{t-1})$ for every t, h_{t-1}.

Fix $m \in \mathbb{N}$, functions

$$c_t : H_t \to \mathbb{R}, \qquad r_t : H_t \to \mathbb{R}^m,$$

and continuous maps

$$\Phi_{t,h_{t-1}} : \mathbb{R}^m \to \mathbb{R} \qquad (t = 1, \dots, T,\ h_{t-1} \in H_{t-1}).$$

For a profile γ, let π^γ be the induced path law. Define, for each t and h_{t-1},

$$J_t(\gamma; h_{t-1}) := \mathbb{E}^{\pi^\gamma}\left[\sum_{u=t}^T c_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right] + \Phi_{t,h_{t-1}}\left(\mathbb{E}^{\pi^\gamma}\left[\sum_{u=t}^T r_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right] \right).$$

For $\eta \in \Gamma_t(h_{t-1})$, let $\gamma[t, h_{t-1} \leftarrow \eta]$ denote the profile obtained by replacing only the kernel $\gamma_t(\cdot, \cdot \mid h_{t-1})$ by η. Call γ a *strong equilibrium* if

$$J_t(\gamma[t, h_{t-1} \leftarrow \eta]; h_{t-1}) \geq J_t(\gamma; h_{t-1})$$

for every t, every history h_{t-1}, and every $\eta \in \Gamma_t(h_{t-1})$.

Finally, define the kernel metric

$$d_{\mathrm{ker}}(\gamma, \tilde{\gamma}) := \max_{1 \leq t \leq T} \max_{h_{t-1} \in H_{t-1}} \left\| \gamma_t(\cdot, \cdot \mid h_{t-1}) - \tilde{\gamma}_t(\cdot, \cdot \mid h_{t-1}) \right\|_1.$$

Theorem 1. *With the notation above:*

(i) ***(Exact backward sweep)*** *Set $A_{T+1} \equiv 0$ and $B_{T+1} \equiv 0$. For $t = T, T-1, \dots, 1$, and for each $h_{t-1} \in H_{t-1}$, let $h_t := (h_{t-1}, x_t, y_t)$ and choose*

$$\gamma_t^*(\cdot, \cdot \mid h_{t-1}) \in \arg\min_{\eta \in \Gamma_t(h_{t-1})} F_{t,h_{t-1}}(\eta; A_{t+1}, B_{t+1}),$$

where

$$F_{t,h_{t-1}}(\eta; A_{t+1}, B_{t+1}) := \sum_{x_t, y_t} \eta(x_t, y_t)\big(c_t(h_t) + A_{t+1}(h_t)\big) + \Phi_{t,h_{t-1}}\left(\sum_{x_t, y_t} \eta(x_t, y_t)\big(r_t(h_t) + B_{t+1}(h_t)\big) \right),$$

and then define

$$A_t(h_{t-1}) := \sum_{x_t, y_t} \gamma_t^*(x_t, y_t \mid h_{t-1})\big(c_t(h_t) + A_{t+1}(h_t)\big),$$

$$B_t(h_{t-1}) := \sum_{x_t, y_t} \gamma_t^*(x_t, y_t \mid h_{t-1})\big(r_t(h_t) + B_{t+1}(h_t)\big).$$

Then the recursion is well-defined; any profile γ^* obtained in this way induces a plan $\pi^* \in \Pi_{\mathrm{bc}}(\mu, \nu)$; and γ^* is a strong equilibrium.

(ii) ***(Inexact backward sweep; cluster points)*** *Let $\varepsilon_n \downarrow 0$. For each n, set $A^{(n)}_{T+1} \equiv 0$, $B^{(n)}_{T+1} \equiv 0$, and recursively choose $\gamma^{(n)}_t(\cdot,\cdot \mid h_{t-1}) \in \Gamma_t(h_{t-1})$ such that*

$$F_{t,h_{t-1}}\Big(\gamma^{(n)}_t(\cdot,\cdot \mid h_{t-1}); A^{(n)}_{t+1}, B^{(n)}_{t+1}\Big) \le \inf_{\eta\in\Gamma_t(h_{t-1})} F_{t,h_{t-1}}(\eta; A^{(n)}_{t+1}, B^{(n)}_{t+1}) + \varepsilon_n,$$

and then define $A^{(n)}_t, B^{(n)}_t$ by the same formulas as above. Let $\pi^{(n)} := \pi^{\gamma^{(n)}}$.

Then $(\gamma^{(n)})_{n\ge1}$ has a subsequence converging in d_{ker}, and every such limit point is a strong equilibrium. Consequently, $(\pi^{(n)})_{n\ge1}$ has total-variation cluster points, all of which are equilibrium bicausal transport plans.

(iii) ***(Full convergence under local uniqueness)*** *If, for every stage t, history h_{t-1}, and continuation pair (A_{t+1}, B_{t+1}) produced by the exact backward recursion, the local problem*

$$\min_{\eta\in\Gamma_t(h_{t-1})} F_{t,h_{t-1}}(\eta; A_{t+1}, B_{t+1})$$

has a unique minimizer, then the backward-recursion profile γ^* is unique, and

$$d_{\mathrm{ker}}(\gamma^{(n)}, \gamma^*) \to 0, \qquad \|\pi^{(n)} - \pi^*\|_{\mathrm{TV}} \to 0.$$

Proof

Lemma 1. *Every kernel profile $\gamma = (\gamma_t)_{t=1}^T$ induces a bicausal plan*

$$\pi^\gamma(x_{1:T}, y_{1:T}) := \prod_{t=1}^T \gamma_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1}),$$

and π^γ has first marginal μ and second marginal ν.

Proof. Because each $\gamma_t(\cdot,\cdot \mid h_{t-1})$ is a probability vector on $X_t \times Y_t$, repeated summation shows that π^γ has total mass 1.

We prove the X-marginal claim by induction on t. For $t = 1$,

$$\sum_{y_1} \pi^\gamma(x_1, y_1) = \sum_{y_1} \gamma_1(x_1, y_1) = \mu_1(x_1).$$

Assume

$$\sum_{y_{1:t-1}} \prod_{u=1}^{t-1} \gamma_u(x_u, y_u \mid x_{1:u-1}, y_{1:u-1}) = \prod_{u=1}^{t-1} \mu_u(x_u \mid x_{1:u-1}).$$

Then

$$\begin{aligned}
&\sum_{y_{1:t}} \prod_{u=1}^{t} \gamma_u(x_u, y_u \mid x_{1:u-1}, y_{1:u-1}) \\
&= \sum_{y_{1:t-1}} \left(\prod_{u=1}^{t-1} \gamma_u(x_u, y_u \mid x_{1:u-1}, y_{1:u-1})\right) \left(\sum_{y_t} \gamma_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1})\right) \\
&= \sum_{y_{1:t-1}} \left(\prod_{u=1}^{t-1} \gamma_u(x_u, y_u \mid x_{1:u-1}, y_{1:u-1})\right) \mu_t(x_t \mid x_{1:t-1}) \\
&= \mu_t(x_t \mid x_{1:t-1}) \prod_{u=1}^{t-1} \mu_u(x_u \mid x_{1:u-1}) = \prod_{u=1}^{t} \mu_u(x_u \mid x_{1:u-1}).
\end{aligned}$$

Thus the X-marginal is μ. The Y-marginal equals ν by the same argument, exchanging the roles of x and y.

It remains to prove bicausality. Fix t. If $\mu_t(x_t \mid x_{1:t-1}) > 0$, define

$$q_t(y_t \mid x_{1:t}, y_{1:t-1}) := \frac{\gamma_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1})}{\mu_t(x_t \mid x_{1:t-1})}.$$

If $\mu_t(x_t \mid x_{1:t-1}) = 0$, define $q_t(\cdot \mid x_{1:t}, y_{1:t-1})$ arbitrarily as a probability vector on Y_t. Since the X-marginal of $\gamma_t(\cdot, \cdot \mid h_{t-1})$ is $\mu_t(\cdot \mid x_{1:t-1})$, we have

$$\sum_{y_t} q_t(y_t \mid x_{1:t}, y_{1:t-1}) = 1$$

whenever the denominator is positive. Then

$$\mu(x_{1:T}) \prod_{t=1}^{T} q_t(y_t \mid x_{1:t}, y_{1:t-1}) = \prod_{t=1}^{T} \gamma_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1}) = \pi^{\gamma}(x_{1:T}, y_{1:T}).$$

So π^{γ} is causal from X to Y.

Similarly, if $\nu_t(y_t \mid y_{1:t-1}) > 0$, define

$$p_t(x_t \mid x_{1:t-1}, y_{1:t}) := \frac{\gamma_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1})}{\nu_t(y_t \mid y_{1:t-1})},$$

and define it arbitrarily otherwise. Then

$$\pi^{\gamma}(x_{1:T}, y_{1:T}) = \nu(y_{1:T}) \prod_{t=1}^{T} p_t(x_t \mid x_{1:t-1}, y_{1:t}),$$

so π^{γ} is causal from Y to X as well. Therefore $\pi^{\gamma} \in \Pi_{\mathrm{bc}}(\mu, \nu)$. $\square$

Lemma 2. *For any kernel profile γ, define recursively*

$$A^{\gamma}_{T+1} \equiv 0, \qquad B^{\gamma}_{T+1} \equiv 0,$$

and for $t = T, \dots, 1$,

$$A^{\gamma}_t(h_{t-1}) := \sum_{x_t, y_t} \gamma_t(x_t, y_t \mid h_{t-1})\big(c_t(h_t) + A^{\gamma}_{t+1}(h_t)\big),$$

$$B^{\gamma}_t(h_{t-1}) := \sum_{x_t, y_t} \gamma_t(x_t, y_t \mid h_{t-1})\big(r_t(h_t) + B^{\gamma}_{t+1}(h_t)\big),$$

where $h_t = (h_{t-1}, x_t, y_t)$.

Then for every t and h_{t-1},

$$A^{\gamma}_t(h_{t-1}) = \mathbb{E}^{\pi^{\gamma}}\left[\sum_{u=t}^{T} c_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right],$$

$$B^{\gamma}_t(h_{t-1}) = \mathbb{E}^{\pi^{\gamma}}\left[\sum_{u=t}^{T} r_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right].$$

Moreover, for any $\eta \in \Gamma_t(h_{t-1})$,

$$J_t(\gamma[t, h_{t-1} \leftarrow \eta]; h_{t-1}) = F_{t,h_{t-1}}(\eta; A^\gamma_{t+1}, B^\gamma_{t+1}).$$

Consequently, γ *is a strong equilibrium if and only if*

$$F_{t,h_{t-1}}\big(\gamma_t(\cdot, \cdot \mid h_{t-1}); A^\gamma_{t+1}, B^\gamma_{t+1}\big) \le F_{t,h_{t-1}}(\eta; A^\gamma_{t+1}, B^\gamma_{t+1})$$

for every $t, h_{t-1}, \eta \in \Gamma_t(h_{t-1})$.

Proof. We prove the identities for A^γ_t and B^γ_t by backward induction on t.

At $t = T$,

$$A^\gamma_T(h_{T-1}) = \sum_{x_T, y_T} \gamma_T(x_T, y_T \mid h_{T-1}) c_T(h_T) = \mathbb{E}^{\pi^\gamma}[c_T(H_T) \mid H_{T-1} = h_{T-1}],$$

and similarly for B^γ_T.

Assume the identities hold at $t + 1$. Then

$$\begin{aligned} A^\gamma_t(h_{t-1}) &= \sum_{x_t, y_t} \gamma_t(x_t, y_t \mid h_{t-1})\big(c_t(h_t) + A^\gamma_{t+1}(h_t)\big) \\ &= \mathbb{E}^{\pi^\gamma}\left[c_t(H_t) + \mathbb{E}^{\pi^\gamma}\left[\sum_{u=t+1}^{T} c_u(H_u) \,\middle|\, H_t \right] \,\middle|\, H_{t-1} = h_{t-1} \right] \\ &= \mathbb{E}^{\pi^\gamma}\left[\sum_{u=t}^{T} c_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right], \end{aligned}$$

where the second equality uses the induction hypothesis, and the third is the tower property. The proof for B^γ_t is identical, componentwise in $\mathbb{R}^m$.

Now fix t, h_{t-1}, and $\eta \in \Gamma_t(h_{t-1})$. Under the deviated profile $\gamma[t, h_{t-1} \leftarrow \eta]$, the law of the immediate successor (X_t, Y_t) conditional on $H_{t-1} = h_{t-1}$ is exactly η, while the future kernels at times $t + 1, \ldots, T$ are unchanged. Therefore

$$\mathbb{E}^{\pi^{\gamma[t,h_{t-1} \leftarrow \eta]}}\left[\sum_{u=t}^{T} c_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right] = \sum_{x_t, y_t} \eta(x_t, y_t)\big(c_t(h_t) + A^\gamma_{t+1}(h_t)\big),$$

and

$$\mathbb{E}^{\pi^{\gamma[t,h_{t-1} \leftarrow \eta]}}\left[\sum_{u=t}^{T} r_u(H_u) \,\middle|\, H_{t-1} = h_{t-1} \right] = \sum_{x_t, y_t} \eta(x_t, y_t)\big(r_t(h_t) + B^\gamma_{t+1}(h_t)\big).$$

Substituting these identities into the definition of J_t gives

$$J_t(\gamma[t, h_{t-1} \leftarrow \eta]; h_{t-1}) = F_{t,h_{t-1}}(\eta; A^\gamma_{t+1}, B^\gamma_{t+1}).$$

The final characterization of strong equilibrium is exactly the same statement rewritten with η arbitrary. $\square$

Lemma 3. *For each* t*, the maps*

$$\gamma \longmapsto A^\gamma_t, \qquad \gamma \longmapsto B^\gamma_t$$

from the compact profile space $\prod_{s,h_{s-1}} \Gamma_s(h_{s-1})$ *into the finite-dimensional spaces of functions on* H_{t-1} *are continuous. Consequently, for each fixed* t, h_{t-1}*, the map*

$$(\eta, \gamma) \longmapsto F_{t,h_{t-1}}(\eta; A^\gamma_{t+1}, B^\gamma_{t+1})$$

is continuous on $\Gamma_t(h_{t-1}) \times \prod_{s,h_{s-1}} \Gamma_s(h_{s-1})$.

Proof. We proceed by backward induction on t. At $t = T+1$, the maps are constant.

Assume continuity holds at $t+1$. Since all history sets are finite,

$$A_t^\gamma(h_{t-1}) = \sum_{x_t, y_t} \gamma_t(x_t, y_t \mid h_{t-1})\big(c_t(h_t) + A_{t+1}^\gamma(h_t)\big)$$

is a finite sum of products of continuous functions of γ, hence is continuous in γ. The same argument applies to $B_t^\gamma(h_{t-1})$. Therefore $\gamma \mapsto A_t^\gamma$ and $\gamma \mapsto B_t^\gamma$ are continuous.

For the last claim, the linear terms in η are continuous, the affine summary

$$\sum_{x_t, y_t} \eta(x_t, y_t)\big(r_t(h_t) + B_{t+1}^\gamma(h_t)\big) \in \mathbb{R}^m$$

is continuous in (η, γ), and $\Phi_{t,h_{t-1}}$ is continuous. Hence $F_{t,h_{t-1}}(\eta; A_{t+1}^\gamma, B_{t+1}^\gamma)$ is continuous jointly in (η, γ). $\square$

Proof of Theorem 1. **Part (i): exact backward sweep.** For fixed t, h_{t-1}, the feasible set $\Gamma_t(h_{t-1})$ is nonempty: for example, the product coupling

$$\eta(x_t, y_t) := \mu_t(x_t \mid x_{1:t-1})\, \nu_t(y_t \mid y_{1:t-1})$$

belongs to it. It is also a closed subset of the finite-dimensional simplex on $X_t \times Y_t$, hence compact. The objective $\eta \mapsto F_{t,h_{t-1}}(\eta; A_{t+1}, B_{t+1})$ is continuous. Therefore a minimizer exists at every stage and history, so the backward recursion is well-defined.

By Lemma 1, the resulting profile γ^* induces a plan $\pi^* \in \Pi_{\mathrm{bc}}(\mu, \nu)$.

To prove strong equilibrium, fix t, h_{t-1}, and $\eta \in \Gamma_t(h_{t-1})$. By construction,

$$F_{t,h_{t-1}}(\gamma_t^*(\cdot,\cdot \mid h_{t-1}); A_{t+1}, B_{t+1}) \le F_{t,h_{t-1}}(\eta; A_{t+1}, B_{t+1}).$$

But, for the exact backward-sweep profile γ^*, the arrays A_{t+1}, B_{t+1} are precisely $A_{t+1}^{\gamma^*}, B_{t+1}^{\gamma^*}$. Therefore Lemma 2 gives

$$J_t(\gamma^*; h_{t-1}) \le J_t(\gamma^*[t, h_{t-1} \leftarrow \eta]; h_{t-1}).$$

Since t, h_{t-1}, η were arbitrary, γ^* is a strong equilibrium.

Part (ii): cluster points of inexact backward sweeps. The full profile space

$$\Gamma := \prod_{t=1}^{T} \prod_{h_{t-1} \in H_{t-1}} \Gamma_t(h_{t-1})$$

is a finite product of compact sets, hence compact. Thus $(\gamma^{(n)})_{n\ge 1}$ has a convergent subsequence in d_{ker}; write $\gamma^{(n_k)} \to \bar{\gamma}$.

Fix t, h_{t-1}, and $\eta \in \Gamma_t(h_{t-1})$. By the definition of the inexact sweep,

$$F_{t,h_{t-1}}\Big(\gamma_t^{(n_k)}(\cdot,\cdot \mid h_{t-1}); A_{t+1}^{(n_k)}, B_{t+1}^{(n_k)}\Big) \le F_{t,h_{t-1}}(\eta; A_{t+1}^{(n_k)}, B_{t+1}^{(n_k)}) + \varepsilon_{n_k}.$$

But, by construction of the sweep,

$$A_{t+1}^{(n_k)} = A_{t+1}^{\gamma^{(n_k)}}, \qquad B_{t+1}^{(n_k)} = B_{t+1}^{\gamma^{(n_k)}}.$$

Since $\gamma^{(n_k)} \to \bar{\gamma}$, Lemma 3 implies

$$A_{t+1}^{\gamma^{(n_k)}} \to A_{t+1}^{\bar{\gamma}}, \qquad B_{t+1}^{\gamma^{(n_k)}} \to B_{t+1}^{\bar{\gamma}}.$$

Again by Lemma 3, both sides of the approximate optimality inequality converge, and $\varepsilon_{n_k} \to 0$. Passing to the limit gives

$$F_{t,h_{t-1}}\big(\bar\gamma_t(\cdot,\cdot \mid h_{t-1}); A^{\bar\gamma}_{t+1}, B^{\bar\gamma}_{t+1}\big) \le F_{t,h_{t-1}}(\eta; A^{\bar\gamma}_{t+1}, B^{\bar\gamma}_{t+1}).$$

Because η was arbitrary, Lemma 2 implies that $\bar\gamma$ is a strong equilibrium.

Thus every convergent subsequence has a strong-equilibrium limit. By Lemma 1, each such limit induces an equilibrium plan in $\Pi_{\mathrm{bc}}(\mu,\nu)$.

Part (iii): full convergence under local uniqueness. Assume every exact local problem along the backward recursion has a unique minimizer. Then the exact backward sweep determines a unique profile γ^*.

Take any convergent subsequence $\gamma^{(n_k)} \to \bar\gamma$. By part (ii), $\bar\gamma$ is a strong equilibrium. At time T, the continuation arrays are $A_{T+1} = 0$, $B_{T+1} = 0$, so the local problem at each h_{T-1} has the unique minimizer $\gamma^*_T(\cdot,\cdot \mid h_{T-1})$. Since $\bar\gamma$ is a strong equilibrium, its terminal kernels must equal these same unique minimizers:

$$\bar\gamma_T(\cdot,\cdot \mid h_{T-1}) = \gamma^*_T(\cdot,\cdot \mid h_{T-1}) \qquad \forall h_{T-1} \in H_{T-1}.$$

Hence $A^{\bar\gamma}_T = A^{\gamma^*}_T$ and $B^{\bar\gamma}_T = B^{\gamma^*}_T$. Repeating the same argument backward for $t = T-1, T-2, \dots, 1$, we get

$$\bar\gamma_t(\cdot,\cdot \mid h_{t-1}) = \gamma^*_t(\cdot,\cdot \mid h_{t-1}) \qquad \forall t,\ h_{t-1}.$$

Thus $\bar\gamma = \gamma^*$.

We have shown that every convergent subsequence of $(\gamma^{(n)})$ converges to the same limit γ^*. Since $(\gamma^{(n)})$ is contained in the compact space Γ, this implies the whole sequence converges:

$$d_{\mathrm{ker}}(\gamma^{(n)}, \gamma^*) \to 0.$$

Finally, for each full path $(x_{1:T}, y_{1:T})$,

$$\pi^{(n)}(x_{1:T}, y_{1:T}) = \prod_{t=1}^{T} \gamma^{(n)}_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1}) \to \prod_{t=1}^{T} \gamma^*_t(x_t, y_t \mid x_{1:t-1}, y_{1:t-1}) = \pi^*(x_{1:T}, y_{1:T}),$$

because it is a finite product of convergent coordinates. There are only finitely many paths, so summing the absolute differences over all paths yields

$$\|\pi^{(n)} - \pi^*\|_{\mathrm{TV}} \to 0.$$

This completes the proof. $\square$

Corollary 1. *Theorem 1 applies to the prompt's nonlinear objective*

$$J(\pi) = \int \Big(\sum_{t=1}^{T} c_t\Big)\, d\pi + G\Big(\int \psi \, d\pi\Big)$$

whenever X_t, Y_t are finite, $\psi : H_T \to \mathbb{R}^m$, and $G : \mathbb{R}^m \to \mathbb{R}$ is continuous: simply take

$$r_T = \psi, \qquad r_t \equiv 0 \ \ (t < T), \qquad \Phi_{t,h_{t-1}} \equiv G.$$

It also covers state-dependent preferences by allowing $\Phi_{t,h_{t-1}}$ to depend on h_{t-1}.

Proof. For the nonlinear objective, the current self's continuation criterion is

$$\mathbb{E}\left[\sum_{u=t}^{T} c_u(H_u) \,\middle|\, H_{t-1} = h_{t-1}\right] + G(\mathbb{E}[\psi(H_T) \mid H_{t-1} = h_{t-1}]) .$$

This is exactly the theorem with $r_T = \psi$ and $r_t = 0$ for $t < T$. State-dependent preferences are already built into the dependence of $\Phi_{t,h_{t-1}}$ on the realized history. $\square$

Gap to the full statement

What remains unproved relative to the full prompt is precise:

- **Continuous spaces.** I only treat finite X_t, Y_t. I do not prove measurable-selection or discretization-consistency results in general Polish spaces.
- **General nonseparable f-divergence regularization.** The theorem covers nonlinear/state-dependent criteria that depend on the future only through a finite-dimensional expected summary vector. A genuinely pathwise nonseparable f-divergence typically cannot be compressed into such a finite summary.
- **Specific inner solvers.** The outer convergence theorem assumes each local subproblem is solved to vanishing global tolerance ε_n. I do not analyze projected gradient, mirror descent, interior-point, neural approximation, or fitted value iteration as the *inner* solver.
- **Rates/sample complexity.** I prove convergence but not explicit rates, and I do not prove statistical/sample-complexity bounds for learned approximations.

So this note gives a rigorous algorithmic theorem for the fully discrete nonlinear/state-dependent equilibrium problem, but not yet for the full Polish/f-divergence generality of the prompt.

Self red-team

- **Counterexample attempt 1: uniqueness omitted.** Take $T = 1$, $X_1 = Y_1 = \{0, 1\}$, $\mu_1 = \nu_1 = (1/2, 1/2)$, $c_1 \equiv 0$, $r_1 \equiv 0$, $\Phi_{1,\varnothing} \equiv 0$. Then every coupling in $\Gamma_1(\varnothing)$ is an equilibrium. Hence an approximate algorithm can oscillate forever. This shows that full-sequence convergence really needs a selection rule or a uniqueness assumption.
- **Counterexample attempt 2: try to break Lemma 1.** I checked the smallest nontrivial case $T = 2$ with binary X_t, Y_t. The telescoping marginal identities work exactly because each local action is a coupling of the *prescribed* one-step marginals. So the feasible deviations are correctly localized.
- **Counterexample attempt 3: try to break the continuity argument.** Because all spaces are finite, every continuation array is given by a finite algebraic expression in the kernel coordinates. So there is no hidden measurability or lower-semicontinuity issue in the proof. This is precisely why the theorem is stated only in the finite case.
- **Three plausible referee objections.**
  1. "Do stagewise couplings really characterize bicausal plans?"
     Yes; Lemma 1 gives an explicit causal disintegration in both directions.
  2. "Why does the nonlinear term still admit a backward recursion?"
     Because future selves are frozen in a one-step deviation. Hence the current deviation affects the future only through the distribution of the immediate successor history, summarized by A_{t+1} and B_{t+1}; this is exactly Lemma 2.
  3. "Approximate minimizers of changing objectives need not converge."
     Correct in general, but here the profile space is compact and the local objectives depend continuously on the profile; therefore every cluster point satisfies the exact stagewise

inequalities. Full convergence needs local uniqueness, and I stated that assumption explicitly.

- **Boundary checks.**
  1. $T = 1$: the theorem reduces to a single optimization over one transport polytope.
  2. $\Phi_{t,h} \equiv 0$: it reduces to the time-consistent additive case.
  3. $r_t \equiv 0$: same reduction.
  4. Histories with zero conditional probability under μ or ν: kernels there can be defined arbitrarily without changing the induced plan, and the proof still goes through because the construction is stronger than the prompt's a.e.-history requirement.

Ideas tried

1. **Successful line: finite-summary backward recursion.**
   Idea: replace the missing Bellman scalar value by a pair (A_t, B_t) of continuation summaries.
   Outcome: this succeeds in the finite-state case and yields Theorem 1.
   Natural next step: extend the same idea to Polish spaces using measurable selection and weak continuity.

2. **Attempted extension to general pathwise f-divergence.**
   Idea: treat the divergence contribution as an extra state variable, analogous to B_t.
   Obstruction: a general f-divergence depends on the whole future conditional law, not on finitely many expected statistics; I could not compress it into a finite-dimensional sufficient state without imposing much stronger structure.
   Natural next step: look for special divergence families admitting a recursive dual representation.

3. **Attempted global best-response fixed-point iteration.**
   Idea: define a synchronous map "current profile $\mapsto$ stagewise best responses" and prove contraction.
   Obstruction: without strong regularization or monotonicity, the map need not be contractive; the $T = 1$, zero-cost example above already shows nonuniqueness and possible oscillation.
   Natural next step: add explicit strongly convex local regularization and seek a Banach-contraction theorem.

4. **Attempted general Polish-space theorem.**
   Idea: use compactness/tightness of kernels, Berge-type maximum arguments, and measurable selectors.
   Obstruction: the proof would require a careful audit of measurable argmin selection and continuity of continuation functionals under weak convergence at every history; I did not obtain a self-contained proof within this note.
   Natural next step: impose compact state spaces and Feller-type continuity, then build the measurable-selection argument explicitly.

Invoked results

No external theorem is used as a logical step in the proof. The argument is self-contained and relies only on finite-dimensional compactness and the three lemmas proved in-line above. The web-searched papers are used only for orientation and context, not as imported proof steps.

Self-confidence

3. I checked the finite-state recursion carefully and could not break any step, but the subject is subtle enough that I prefer to reserve a small margin for hidden interpretation issues about equilibrium on off-path histories.